

to make better treatment decisions. XAI can also be used to identify biases in the data or the model, which can help to improve the accuracy and fairness of the model's predictions.

Generative AI

Generative AI is at the forefront of creative innovation. Through neural networks, this technology enables the generation of content, art, music, and even human-like text. Artists, designers, and authors are using generative AI to create one-of-a-kind and inventive works. AI-powered technologies aid in the creation of unique designs in the field of design, while generative AI composes tunes that fascinate listeners in the field of music. However, ethical concerns arise due to its potential for generating deepfake content and misinformation.

In recent years, generative AI has made considerable advances, allowing it to create highly realistic and imaginative content across a wide range of areas. The invention of generative adversarial networks (GANs), a type of machine learning model, is one of the most breakthrough advances in generative AI. GANs are made up of two neural networks: a generator and a discriminator. The generator generates new data instances, while the discriminator validates them. These two networks compete in a process in which the generator wants to create data that is indistinguishable from genuine data and the discriminator aims to correctly categorise real and generated data. The following are some of the most recent advances in Generative AI.

High-quality image synthesis: GANs can generate high-resolution images that often resemble real photographs. This can be used to create art, design, and even photorealistic representations of non-existent things or landscapes.

Style transfer and artistic rendering: Generative AI allows users to apply one image's style to another's content. AI-generated art has emerged

as a result, combining the qualities of many styles and materials.

Text-to-image synthesis: Generative AI can now create visuals that closely match the specified scenarios when given a textual cue, making it useful in sectors such as creative writing and advertising.

Video generation and manipulation: Advances in GANs simplify video generation and manipulation, enabling AI to produce new video content, alter current video footage, and even interpolate between frames to produce seamless transitions.

Voice and audio synthesis: Beyond images, Generative AI expands into audio synthesis, creating lifelike speech, music, and sound effects for entertainment, gaming, and voice assistants.

Drug discovery and molecule generation: Generative AI aids in identifying novel medicinal compounds by producing chemical structures that match specified criteria. This speeds up the drug discovery process by allowing researchers to investigate a broader range of possible molecules.

Realistic human face generation: AI models create incredibly convincing human faces, raising ethical issues about future applications in deepfake content creation.

AI-generated content curation: Generative AI supports content creators by suggesting ideas, generating drafts, and even composing music or text segments depending on specific topics or inputs.

Cross-domain applications:

Generative AI finds applications across domains, allowing artists, designers, scientists, and engineers to use its creative powers in a variety of applications.

As generative AI advances, it has the potential to transform businesses, ignite creativity, and provide new ways to engage with and produce material. However, ethical concerns, such as the possibility of misuse and the need to ensure the responsible use of AI-generated content, must be addressed as this technology advances.

AI fairness and bias

Famous AI guru Ray Kurzweil coined the term 'singularity' in artificial intelligence, signifying the convergence of AI and human intelligence or making AIML accuracy approach human-like thinking. To achieve this highest level of accuracy in ML training, modelling, and functionality, it is of the utmost importance to address fairness and bias in AIML implementation.

Efforts to reduce or avoid bias in developing an AIML solution across various stages naturally lead to fairness in execution and results. Bias cannot occur on its own in the AIML stages by the system, but can result from human inputs.

For example, when preparing data for ML model training, one should have fairly distributed, unbiased data to prevent biased outcomes. Similarly, when labelling and grouping the data, training the model to simulate human-like thinking, deploying the model with knowledge gained from training, and interpreting the results from ML execution, it is essential to avoid pre-judgement or biased personal interests that could compromise fairness in building the entire system.

This includes avoiding race, income category, sexual orientation, gender, religion, marginalisation of the user and any discrimination of the user data when preparing the training data, as well as training the system and interpreting the results from the machine learning execution.

Various tools like IBM AI Fairness 360, Microsoft Fairlearn, and Google What-if are invaluable for validating and identifying if the training model and data collection has any bias in preparation and implementing fairness in the AI applications.

AI governance

AI governance is a set of principles, policies, and practices that organisations use to manage the development, deployment, and use of AI systems. It is important to have AI